

B. Sc. Fifth Semester
DSC – 5C: Chemistry Paper–V

UNIT – I

12 Hours

Acids and Bases:

02 Hours

Definition of acids and bases, Arrhenius theory, Lowry-Bronsted theory, Lewis concept and Lux-Flood theory. HSAB principle and applications

Non-aqueous Solvents:

03Hours

Introduction, physical properties of a solvent, types of solvents and their general characteristics. Reaction in nonaqueous solvents with reference to liquid ammonia, liquid sulphur dioxide, liquid H_2SO_4 and N_2O_4 .

Metallurgy

07 Hours

Introduction: Ore, minerals, important ores of some common elements in India, General Principles of pyrometallurgy, roasting, Calcination, Gangue, Smelting, Flux, Gravity separation, Froth flotation process, leaching. Techniques employed for Purification of metal (Distillation process, Bessemerization, Electro-refining, Van Arkel and De Boer's Filament).

Extraction of metals: Extraction of Manganese (Pyrolusite), Titanium (Ilmenite) and Uranium.

UNIT-II

12 Hours

Carbohydrates:

06 Hours

Introduction and classification. Mechanism of osazone formation, Inter conversion of Glucose into Fructose and vice-versa. Chain lengthening of aldoses (Killiani-Fischer synthesis), chain shortening of aldoses (Wohl degradation). Epimerization and mutarotation. Elucidation of structure of glucose (open chain structure and configuration).

Determination of ring size in glucose (Fischer and Haworth structure of glucose). Haworth structure of sucrose, maltose, lactose

Amino Acids, peptides and proteins:

06 Hours

Introduction, classification and structure of amino acids, Synthesis of α – amino acids (from acids, Strecker synthesis and Gabriel- phthalamide synthesis). Acid-Base behavior and isoelectric point of an amino acid.

Peptides-classification, peptide linkage. Synthesis of a dipeptide (Glycylalanine). Proteins – Classification of proteins on composition and molecular shape. Primary and secondary structures of proteins.

UNIT-III

12 Hours

Electrochemistry –I

08 Hours

Revision of conduction in metals and electrolyte solutions, Specific conductance, molar conductance, equivalent conductance, variation of equivalent conductance and specific conductance with dilution. Cell constant, determination of equivalent conductance. Ionic conductance, ionic mobility, Kohlrausch's and its applications. Numerical problems.

Application of conductance measurements

- a) Solubility and solubility product of sparingly soluble salt
- b) Degree of dissociation of weak electrolytes
- c) Conductometric titrations: Acid-base and precipitation titrations and advantages of these.

Phase Equilibria:

04 Hours

Phase rule, definition and explanation of the terms involved in the phase rule. Application to one component system of water, triple point, application to two component system (lead- silver and potassium iodide- water system), Eutectic mixtures and their applications. Freezing mixtures.

UNIT – IV

12 Hours

Spectroscopy:

12 Hours

Introduction, Electromagnetic radiation. Principle, instrumentation and applications of UV-VIS, FT-IR, NMR spectroscopy in organic analysis. Meaning of the terms chromophore, auxochromophore, bathochromic and hypsochromophoric shifts with examples. Functional, Equivalent and non-equivalent protons, chemical shift, spin-spin coupling, (n+1) rule, Shielding and de-shielding of protons in NMR spectroscopy

Chemistry Paper–V: Blue Print

Sl. No.	Name of the chapter	No. of hours	Marks
1	Acids and Bases	02	03
2	Non-aqueous Solvents	03	05
3	Metallurgy	07	12
4	Carbohydrates	06	10
5	Amino Acids, peptides and proteins:	06	10
6	Electrochemistry –I	08	13
7	Phase Equilibria	04	07
8	Spectroscopy	12	20
Total		48	80

Organic Chemistry:

Qualitative analysis of binary mixture containing two solid compounds, separation using NaHCO_3 , NaOH and HCl . Identification of mixture, separation of mixture and analysis of any one component with preparation of derivative.

- Acids: Benzoic, Salicylic, Cinnamic and phthalic acid.
- Phenols: 1-Naphthol, 2-Naphthol and resorcinol.
- Bases: p-Toluidine, o-Toluidine, m-Toluidine, Nitroanilines
- Neutral: Naphthalene, Diphenyl, m-dinitrobenzene, Acetanilide.

The mixture is of A+B, A+N, P+B, P+N and B+N

Inorganic Chemistry:

1. Separation of metal ions of group IV by ascending chromatography.
2. Separation of metal ions of group II by paper chromatography.
3. Separation of Phenols on activated silica gel (TLC).

Note: Qualitative analysis given in the examination.

Physical Chemistry:**Instrumental methods of analysis-I**

1. Determination of degree of dissociation of weak electrolyte.
2. Determination of dissociation constant (K_a) of weak electrolyte.
3. Determination of equivalent conductance at infinite dilution of strong electrolyte.
4. Determination of solubility and solubility product of a sparingly soluble salt (say BaSO_4 , AgCl , AgBr) by conductance method.
5. Determine the amount of Iron in FeCl_3 solution and verify the Beer-Lambert's law.
6. Prepare a standard 0.2M N_2HPO_4 and 0.1M citric acid solution, hence prepare Different buffer solutions and determine K_a values of these unknown solutions.
7. Determine the $P K_a$ value as of given acid by pH metric titration with strong base.
8. Determine the percentage of cane sugar or tartaric acid solution using polarimeter.